Scenario: "Walk Me Through What Happens When You Type a URL and See a React Page"

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10.4 - Scenario: "Walk Me Through What Happens When You Type a URL and See a React Page"

🎯 Why This Question Exists

This tests your understanding of the full stack: networking, server, browser, bundling, React initialization, and rendering.

The Complete Sequence

1. DNS Resolution

The browser looks up the domain name and resolves it to an IP address (checking browser cache, OS cache, router cache, then DNS server).

2. TCP/TLS Connection

The browser establishes a TCP connection (3-way handshake). For HTTPS, a TLS handshake follows for encryption.

3. HTTP Request

The browser sends an HTTP GET request with headers (Accept, User-Agent, cookies, Authorization).

4. Server Response

For a React SPA (client-side rendered), the server responds with a minimal HTML file containing a `<div id="root"></div>` and `<script>` tags pointing to the JS bundle. For SSR (Next.js), the server runs React on the server and sends pre-rendered HTML.

5. Browser Parses HTML

The browser constructs the DOM. When it encounters `<script>` tags, it downloads the JavaScript bundles.

6. JavaScript Executes

React's entry point runs:

JSX Copy

```jsx
const root = createRoot(document.getElementById('root'));
root.render(<App />);
```

7. React Builds the Virtual DOM

React calls your component functions recursively down the tree, creating a Virtual DOM (a JavaScript object tree describing the UI).

8. React Commits to the Real DOM

React creates real DOM nodes from the Virtual DOM and inserts them into `<div id="root">`.

9. Effects Run

After painting, `useEffect` callbacks execute. Data fetching, event listener setup, and side effects happen here.

10. React Router Matches the URL

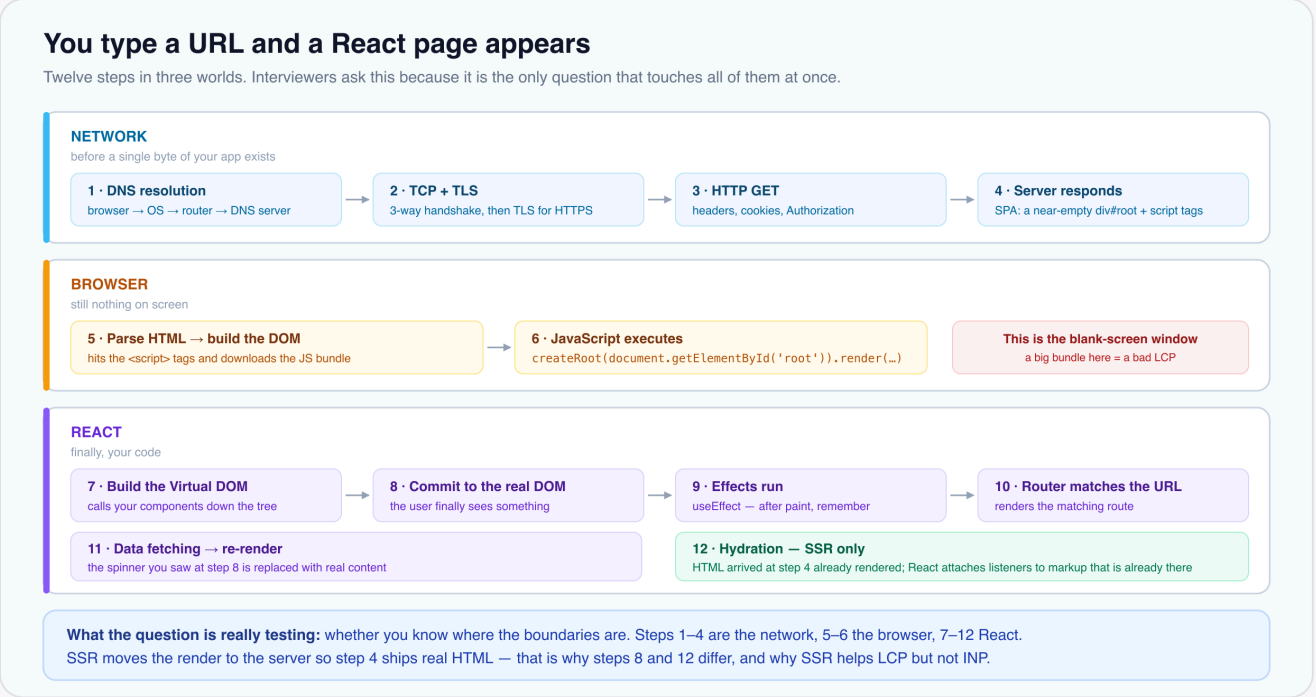
React Router checks the current URL against defined routes and renders the matching component. For `/products/42`, it matches `<Route path="/products/:id">` and provides `{ id: '42' }` via `useParams()`.

11. Data Fetching and Re-render

The matched component fires API calls. Loading states show while data is fetched. Once data arrives, state updates trigger a re-render with the actual content.

12. Hydration (SSR only)

If the page was server-rendered, React "hydrates" the existing HTML by attaching event listeners and state. The page was already visible but becomes fully interactive after hydration.



Sample Answer

> "When you type a URL, the browser resolves the domain via DNS, establishes a TCP/TLS connection, and sends an HTTP GET request. For a React SPA, the server

returns a minimal HTML shell with script tags. The browser parses the HTML, downloads the JS bundle, and React's `createRoot` mounts the App component.

React builds a Virtual DOM, diffs it against the empty DOM, and commits real DOM nodes. After painting, useEffect callbacks run for data fetching. React Router matches the URL to the correct route component. If it's SSR with Next.js, the server pre-renders full HTML so users see content immediately, and React hydrates the page on the client to make it interactive."

QUICK CHECK — no marks, just for you.

1. For a purely client-side-rendered React SPA, what does the server send back in response to the initial HTTP GET request for the page?

- ◯ The compiled JS bundle only, with no HTML document at all
- ◯ A JSON payload describing the Virtual DOM, which the browser turns into elements
- ◯ Fully pre-rendered HTML for the matched route, produced by running React on the server
- ◯ A minimal HTML shell with a root div and script tags pointing to the JS bundle

Check answer

2. At what point in the sequence do useEffect callbacks (for example, the component's initial data fetch) actually execute?

- ◯ Only after React Router has finished matching the URL and the API response arrives
- ◯ While React is recursively building the Virtual DOM tree, before any commit
- ◯ Immediately after createRoot runs, before the App component is called
- ◯ After React commits to the real DOM and the browser paints

3. On an SSR page built with Next.js, what does hydration change for the user?

- ○ Nothing is visible until hydration finishes, because React must create the DOM nodes first
- ○ It replaces the server HTML with freshly created DOM nodes built from the Virtual DOM
- ○ The content was already visible; hydration attaches event listeners and state so it becomes interactive
- ○ It re-runs DNS resolution and the TCP/TLS handshake so the client can fetch the JS bundle

Check answer

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